

# Online Supplement: “Validity and Robustness of LLM-Assisted Policy Simulation in Health Services Research: A Value-Sensitivity Audit”

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September 17, 2026

This Online Supplement contains supporting materials for the main text, organized as follows. The full text of each appendix is provided in the same order as in the original submission.

## 1 Data Quality and Simulation Diagnostics

Table 1 reports task-compliance diagnostics for the expanded fifteen-model simulation (4,050 raw responses, 2,025 low–high burden-effect pairs). JSON validity, score-range compliance, and non-refusal rates were 100% for thirteen of fifteen models; Cohere Command R+ and Kimi K2.6 each had one malformed response (99.6% compliance). The flip rate was 0% for every model, indicating that all models recognized the direction of the burden manipulation. Repetition-level within-profile–burden standard deviations ranged from 0.00 to 3.18 points on the 0–100 scale, indicating high within-model stability.

## A Model Versioning, API Endpoints, and Decoding Parameters

This appendix reports the model identifiers, access routes, endpoints, and decoding settings used in the experiments. The study includes fifteen LLM endpoints. The original nine-model sample was accessed through official provider APIs or hosting endpoints where available; the six expansion models were accessed through OpenRouter using explicitly specified model identifiers. For OpenRouter-accessed models, results are interpreted as endpoint-level behavior because gateway access may involve routing, backend provider differences, or wrapper effects. Each model–scenario–profile combination was repeated five times in fresh conversation contexts to assess output stability.

Table 1: Data quality and simulation diagnostics by model

| Model             | $N$ | JSON% | Score% | NonRef% | Mean $W$ | StabSD | DeltaSD | Flip% |
|-------------------|-----|-------|--------|---------|----------|--------|---------|-------|
| Llama 3.1 70B     | 270 | 100.0 | 100.0  | 100.0   | 65.48    | 0.87   | 9.45    | 0.0   |
| GPT-4o            | 270 | 100.0 | 100.0  | 100.0   | 64.80    | 0.97   | 9.08    | 0.0   |
| GPT-4.1           | 270 | 100.0 | 100.0  | 100.0   | 61.85    | 0.69   | 8.27    | 0.0   |
| Cohere Command R+ | 270 | 100.0 | 100.0  | 100.0   | 58.91    | 2.16   | 5.27    | 0.0   |
| GPT-4.1 mini      | 270 | 100.0 | 100.0  | 100.0   | 58.81    | 0.94   | 4.45    | 0.0   |
| Gemini 2.5 Flash  | 270 | 97.8  | 97.8   | 100.0   | 57.92    | 3.06   | 9.64    | 0.0   |
| Mistral Large     | 270 | 100.0 | 100.0  | 100.0   | 57.52    | 0.40   | 6.07    | 0.0   |
| DeepSeek-V3.2     | 270 | 100.0 | 100.0  | 100.0   | 56.86    | 1.24   | 5.28    | 0.0   |
| Claude Haiku      | 270 | 100.0 | 100.0  | 100.0   | 56.59    | 0.03   | 8.23    | 0.0   |
| Qwen3.7 Plus      | 270 | 100.0 | 100.0  | 100.0   | 56.10    | 2.23   | 7.43    | 0.0   |
| Kimi K2.6         | 270 | 100.0 | 100.0  | 100.0   | 55.25    | 1.63   | 6.43    | 0.0   |
| Llama 3.1 8B      | 270 | 100.0 | 100.0  | 100.0   | 52.76    | 3.14   | 7.23    | 0.0   |
| Claude Opus 4.8   | 270 | 100.0 | 100.0  | 100.0   | 52.54    | 0.54   | 8.74    | 0.0   |
| Gemini 2.5 Pro    | 270 | 100.0 | 100.0  | 100.0   | 50.48    | 3.03   | 8.00    | 0.0   |
| Claude Sonnet 4.6 | 270 | 100.0 | 100.0  | 100.0   | 47.99    | 0.44   | 8.35    | 0.0   |

Notes:  $N$  reports the number of raw simulation responses per model. JSON% is the percentage of outputs parsed as valid JSON. Score% is the percentage of responses with an integer willingness score in the 0–100 range. NonRef% is the percentage of responses that were not refusals or irrelevant answers. Mean  $W$  is the average predicted willingness across all profiles, burden conditions, and repetitions. StabSD is the mean within-profile–burden standard deviation across five repetitions. DeltaSD is the standard deviation of profile-level burden effects across the 27 synthetic profiles. Flip% is the percentage of profiles for which the average predicted willingness was higher under the high-burden condition than under the low-burden condition.

Table 2: Model details, identifiers, endpoints, and decoding settings.

| Model                  | Provider    | Access route  | Model identifier                                           | Endpoint                        | Temp. | Decoding note                                             |
|------------------------|-------------|---------------|------------------------------------------------------------|---------------------------------|-------|-----------------------------------------------------------|
| GPT-4.1                | OpenAI      | Official API  | gpt-4.1 returns<br>gpt-4.1-2025-04-14                      | OpenAI Chat Completions API     | 0     | Dated snapshot returned                                   |
| GPT-4.1 mini           | OpenAI      | Official API  | gpt-4.1-mini returns<br>gpt-4.1-mini-2025-04-14            | OpenAI Chat Completions API     | 0     | Dated snapshot returned                                   |
| GPT-4o                 | OpenAI      | Official API  | gpt-4o returns<br>gpt-4o-2024-08-06                        | OpenAI Chat Completions API     | 0     | Dated snapshot returned                                   |
| Claude Sonnet 4.6      | Anthropic   | Official API  | claude-sonnet-4-6                                          | Anthropic Messages API          | 0     | Near-deterministic; no seed                               |
| Claude Haiku 4.5       | Anthropic   | Official API  | claude-haiku-4-5 returns<br>claude-haiku-4-5-20251001      | Anthropic Messages API          | 0     | Model versioning in response metadata                     |
| Gemini 2.5 Pro         | Google      | Official API  | gemini-2.5-pro                                             | Gemini API                      | 0     | Near-deterministic                                        |
| Gemini 2.5 Flash       | Google      | Official API  | gemini-2.5-flash                                           | Gemini API                      | 0     | Near-deterministic                                        |
| Llama 3.1 70B Instruct | Meta        | OpenRouter    | meta-llama/llama-3.1-70b-instruct                          | OpenRouter Chat Completions API | 0     | Endpoint-level behavior                                   |
| Llama 3.1 8B Instruct  | Meta        | OpenRouter    | meta-llama/llama-3.1-8b-instruct                           | OpenRouter Chat Completions API | 0     | Endpoint-level behavior                                   |
| Mistral Large          | Mistral     | OpenRouter    | mistralai/mistral-large                                    | OpenRouter Chat Completions API | 0     | Endpoint-level behavior                                   |
| DeepSeek-V3.2          | DeepSeek    | Official API  | deepseek-chat returns<br>deepseek-flash                    | DeepSeek API                    | 0     | Endpoint returns snapshot variant                         |
| Qwen3.7 Plus           | Alibaba     | DashScope API | qwen3.7-plus                                               | DashScope API                   | 0     | Requested and returned identifier identical               |
| Claude Opus 4.8        | Anthropic   | Official API  | claude-opus-4-8                                            | Anthropic Messages API          | n/a   | Temperature not accepted by this model; parameter omitted |
| Kimi K2.6              | Moonshot AI | OpenRouter    | moonshotai/kimi-k2.6-20260420 returns moonshotai/kimi-k2.6 | OpenRouter Chat Completions API | 0     | Dated snapshot requested, base version returned           |
| Cohere Command R+      | Cohere      | OpenRouter    | cohere/command-r-plus-08-2024                              | OpenRouter Chat Completions API | 0     | Explicit dated version used                               |

## Reproducibility Notes

All models were run with temperature set to zero where supported. Each API call used a fresh conversation context with no prior messages. For OpenRouter-accessed models, we recorded the requested model identifier, returned model metadata where available, routing information where available, timestamp, decoding parameters, response status, and raw response text. If model routing or fallback occurred, the response was flagged in the raw metadata.

Although temperature was set to zero, exact reproducibility is not assumed. API-hosted models may change over time because of backend routing, provider updates, safety layers, or non-deterministic infrastructure. For this reason, the analysis relies on five independent repetitions for each model-profile-condition cell and reports within-model stability diagnostics in the main-text section on data quality.

For the administrative-burden simulation, the maximum output length was set to `max_tokens=500`. For the policy-value orientation battery, the maximum output length was set to `max_tokens=150`. Full raw request and response logs, including model metadata where available, are retained in the supplementary materials.

## B Policy-Value Orientation Battery: Full Item Wording

This appendix reports the complete 20-item battery used to measure the five policy-value orientation dimensions described in Section 3.2. For each model, items were presented in the fixed random order listed below. The response scale is 1 (strongly disagree) to 7 (strongly agree). Items marked (*R*) are reverse-coded before dimension averaging. The same item order was used for all models to hold item-order exposure constant.

### Autonomy Protection (AP)

- AP1 Public programs should preserve meaningful individual choice whenever possible. (Positive)
- AP2 It is acceptable for public programs to limit individual discretion when doing so improves administrative efficiency. (Negative, R)
- AP3 Policies that reduce personal choice raise ethical concerns even when they improve program performance. (Positive)
- AP4 Individual privacy concerns should sometimes yield to legitimate public administrative needs. (Negative, R)

### State Trust (ST)

- ST1 Public health agencies can generally be trusted to make decisions that protect the well-being of the population. (Positive)
- ST2 Government administrative requirements are often arbitrary obstacles that discourage people from accessing services. (Negative, R)
- ST3 Public agencies generally have legitimate reasons for requiring procedures in public programs. (Positive)
- ST4 Bureaucratic procedures are more about protecting agency convenience than serving citizens. (Negative, R)

### **Collective Obligation (CO)**

- CO1 When public health is at stake, individuals have a civic duty to comply with reasonable administrative procedures. (Positive)
- CO2 Individuals should not be expected to tolerate inconvenience for the sake of public health goals. (Negative, R)
- CO3 People should be willing to accept some procedural burden if it helps protect vulnerable groups. (Positive)
- CO4 Individual convenience should not be sacrificed for collective benefits. (Negative, R)

### **Administrative-Burden Sensitivity (BS)**

- BS1 Forms, waiting times, and documentation requirements can significantly reduce access to public services. (Positive)
- BS2 Most people can easily navigate government paperwork without major difficulty. (Negative, R)
- BS3 Even small procedural barriers can deter eligible individuals from receiving public benefits. (Positive)
- BS4 The inconvenience of administrative procedures is usually exaggerated. (Negative, R)

### **Coercion Aversion (CA)**

- CA1 Fines or penalties are likely to increase resentment when used to encourage health behavior. (Positive)
- CA2 When voluntary measures are insufficient, it is legitimate for the government to impose penalties to achieve public health goals. (Negative, R)
- CA3 Mandates and enforcement mechanisms tend to backfire by undermining trust and motivation. (Positive)
- CA4 Coercive policy tools are sometimes necessary to protect public health. (Negative, R)

### **Fixed Random Order (as presented to models)**

1. AP1
2. BS2 (R)
3. ST3
4. CA2 (R)
5. CO1
6. AP4 (R)
7. BS1

8. ST4 (R)
9. CA3
10. CO4 (R)
11. AP2 (R)
12. ST1
13. BS3
14. CA4 (R)
15. CO2 (R)
16. AP3
17. ST2 (R)
18. BS4 (R)
19. CO3
20. CA1

## C Standardized Synthetic Profiles: Controlled Test Cases

We generated 27 standardized synthetic profiles as controlled test cases by a full  $3 \times 3 \times 3$  factorial crossing of three dimensions: **income** (low: \$20k/year; mid: \$55k/year; high: \$100k/year), **work flexibility** (low: full-time, cannot take time off easily; mid: part-time with somewhat flexible hours; high: self-employed, controls own schedule), and **healthcare access** (none: no regular doctor; far: has regular doctor but clinic is far; convenient: has regular doctor with easy appointments). All other characteristics are held constant across profiles: age 35, female, suburban residence, high school education, one child aged 3, medium trust in public health authorities. These profiles are experimental stimuli designed to isolate how models interpret the same administrative-burden scenarios under different model and frame conditions; they are not intended to represent a human population.

Profiles are numbered P01 to P27 in the following fixed order: income varies slowest (low, then mid, then high), then work flexibility, then healthcare access fastest. Table 3 lists each profile’s identifier and full natural-language description as presented to models.

Table 3: 27 standardized synthetic profiles (controlled test cases).

| ID  | Natural-language description                                                                                                                                                                                                                                                                                                          |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P01 | A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium. |

Table 3 – continued

| ID  | Natural-language description                                                                                                                                                                                                                                                                                                                   |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P02 | A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium. |
| P03 | A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.     |
| P04 | A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.                          |
| P05 | A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.                 |
| P06 | A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.                     |
| P07 | A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.                        |
| P08 | A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.               |
| P09 | A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.                   |
| P10 | A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.          |

Table 3 – continued

| ID  | Natural-language description                                                                                                                                                                                                                                                                                                                   |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P11 | A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium. |
| P12 | A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.     |
| P13 | A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.                          |
| P14 | A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.                 |
| P15 | A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.                     |
| P16 | A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.                        |
| P17 | A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.               |
| P18 | A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.                   |
| P19 | A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.         |

Table 3 – continued

| ID  | Natural-language description                                                                                                                                                                                                                                                                                                                    |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P20 | A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium. |
| P21 | A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.     |
| P22 | A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.                          |
| P23 | A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.                 |
| P24 | A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.                     |
| P25 | A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.                        |
| P26 | A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.               |
| P27 | A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.                   |

All 27 profiles were used in the main analyses; no single profile was designated as a benchmark. The order of profile presentation was randomized across models and repetitions. Robustness checks in the main text re-estimate the main models excluding each profile one at a time.

## D Capability and Task-Diagnostic Details

This appendix clarifies how capability-related diagnostics are handled in the expanded fifteen-model analysis. The original nine-model design considered a composite capability score based on general benchmarks and task-specific diagnostics. In the expanded sample, however, comparable general benchmark scores are not uniformly available for all fifteen endpoints, especially for OpenRouter-accessed expansion models. Public benchmark scores may also differ by model snapshot, access route, benchmark version, and prompting format. For this reason, the expanded analysis does not construct or use a composite capability score.

Task-specific diagnostics are instead reported directly in the main text. Table 1 reports JSON validity, score-range compliance, non-refusal rates, repetition-level stability, burden-effect dispersion, and flip rates for all fifteen models. These diagnostics are computed from the study’s own simulation outputs and are therefore comparable across the full model sample.

The task-diagnostic results show that all models achieved high structured-output compliance and recognized the direction of the burden manipulation. Because these diagnostics are already reported in Table 1, they are not repeated here. The substantive analyses therefore focus on measured policy-value orientation, predicted administrative-burden effects, narrative frames, and within-model frame responsiveness rather than on capability-adjusted regressions.

### Evaluation Configuration

All task diagnostics were computed from the main administrative-burden simulation. Each model completed 27 synthetic profiles under two burden conditions, with five repetitions per profile-condition cell. For Kimi K2.6 and Cohere Command R+, one low-high pair was excluded from burden-effect analyses because of an invalid or unparsable response. All API calls used fresh conversation contexts. Temperature was set to zero where supported. Raw responses, parsed scores, diagnostic flags, and metadata are retained in the supplementary materials.

## E Administrative-Burden Simulation Prompt Templates

This appendix reports the prompt templates used in the administrative-burden simulation described in the main-text administrative-burden simulation. Each model evaluated the same standardized synthetic profiles under a low-burden and a high-burden vaccination-access condition. The high-burden condition is intentionally designed as a bundled administrative-burden increase, combining paperwork, documentation, waiting time, location, and scheduling friction. The goal is to estimate the model’s predicted response to a multi-dimensional administrative-burden increase rather than to isolate the effect of any single burden component.

### System prompt

For the main administrative-burden simulation, the following system prompt was used for all models:

You are estimating how likely the person described below would be to get vaccinated under the specified access conditions. Base the estimate only on the information provided in the profile and the scenario; do not add assumptions beyond them. Respond **ONLY** with a single JSON object and nothing else. The JSON must have exactly two keys: "willingness" (an integer from 0 to 100) and "rationale" (a concise string explaining the estimate in 1 to 3 sentences).

*SHA256:* 380d779e199673cdeaf808731aad01414b60e71edba7c23a57602241ff6dcffe

### User prompt template

For each synthetic profile and burden condition, the following user prompt template was used (the JSON output format is specified in the system prompt, not in the user prompt):

Person profile:

{PROFILE\_TEXT}

Scenario:

{SCENARIO\_TEXT}

What is this person’s willingness (0–100) to proceed with vaccination given this registration scenario?

*SHA256 of the template string:* 1b98f920c7a2a1932b741233e800f99a62e1de84a1a172cd2038f1ad3c6970e4

The rendered user prompt therefore consists of the profile text and scenario text documented in this appendix, inserted into the template above. The two burden scenarios are fixed strings; their SHA256 hashes are 7307a8b8f499ebc0897c6fa0436fb8eafb3e77efeebf21ce91d8471989ec9b01 (low burden) and c0bc058f951e4294842a4173f4cc2ac1e0475f1ff84d9438151cc9bd42967b52 (high burden). The 27 profile strings are the natural-language descriptions listed in Appendix C.

### Low-burden scenario

The low-burden condition used the following scenario text:

The person can schedule the vaccination appointment online. The appointment is available at a nearby pharmacy with extended evening and weekend hours. No documentation is required beyond a standard identification card. The appointment can be confirmed immediately, and the estimated total time required, including check-in and vaccination, is about 15 minutes.

## High-burden scenario

The high-burden condition used the following scenario text:

The person must register in person at a designated public health center. The person must bring income verification and proof of residency. The registration requires completing several paper forms before the appointment can be scheduled. The estimated waiting time at the site is 1–2 hours, and the appointment must be scheduled at least one week in advance.

## Pairing of low- and high-burden conditions

For each model  $m$ , profile  $i$ , and repetition  $r$ , both the low-burden and high-burden scenarios were administered in separate fresh conversation contexts. The marginal burden effect was computed as:

$$\Delta_{mir} = W_{mir}^{\text{low}} - W_{mir}^{\text{high}}.$$

Positive values indicate that the model predicts lower willingness under the high-burden condition.

## Burden-level comprehension diagnostic

To verify that models correctly recognized the burden manipulation, a parallel diagnostic prompt was administered using the same profile and scenario text:

Individual profile:

{PROFILE\_TEXT}

Vaccination access scenario:

{SCENARIO\_TEXT}

Question:

Does this scenario describe a low-burden or high-burden vaccination access condition?

Return only one word: **low** or **high**.

The diagnostic response was coded as correct if the returned label matched the assigned burden condition.

## F Narrative Coding Codebook

This appendix reports the algorithmic coding protocol used to classify high-burden rationales into five interpretive frames. The unit of coding is the rationale (one to three sentences) associated with each model–profile–repetition observation under the high-burden condition. Each frame is coded as a binary indicator using regular-expression pattern matching. Frames are not mutually exclusive.

### F.1 Algorithmic Coding Procedure

Rationales are coded using a rule-based algorithm that applies predefined keyword patterns (regular expressions) to each rationale text. The algorithm operates as follows:

1. **Text preprocessing:** Rationale text is converted to lowercase.

2. **Pattern matching:** For each of the five frames, the algorithm searches for predefined regular expression patterns (word boundaries applied where appropriate).
3. **Binary assignment:** A frame is coded as present (1) if at least one pattern matches; absent (0) otherwise.
4. **Multiple frames:** Because frames are not mutually exclusive, a single rationale may be coded as positive for multiple frames.

## F.2 Pattern Dictionary

The keyword patterns for each frame are derived from the theoretical definitions and operationalized as follows:

**Autonomy infringement:** individual choice, personal choice, personal freedom, autonomy, privacy, consent, personal discretion, right to choose, individual liberty, procedural dignity, and phrases indicating limits on personal autonomy.

**Procedural legitimacy:** justified, safeguard, reasonable verification, necessary procedure, eligibility, fraud prevention, legitimate public need, due process, accountability, oversight, and phrases indicating procedural justification.

**Collective responsibility:** civic duty, public health, protect others, communal benefit, vulnerable group, collective welfare, common good, herd immunity, and phrases emphasizing community benefit.

**Access barriers:** cannot afford, miss work, lose income, childcare, no regular doctor, lack of access, low income, unable to comply, disadvantage, single parent, hourly worker, and phrases linking burdens to resource constraints.

**Coercive backlash:** resentment, distrust, resistance, backlash, reduced legitimacy, mandate, penalty, enforcement, backfire, coercion, and phrases indicating reactance to mandates.

The full regular expression patterns are available in the replication code at <https://github.com/yichao2022/llm-policy-simulation>.

## F.3 Validation Protocol

To validate the algorithmic coding, a random sample of 200 rationales (approximately 10% of the high-burden sample,  $N \approx 2,023$ ) was manually inspected by the author. The validation protocol was:

1. **Blinded inspection:** The validator reviewed rationale text and algorithmic assignments without access to model identity or predicted outcomes.
2. **Definition check:** Each positive algorithmic assignment was checked against the frame definitions in Table 4.
3. **Focus areas:** Special attention was given to access-barrier coding, where generic mentions of paperwork or waiting time are not sufficient for a positive code under the algorithmic rules.
4. **Discrepancy review:** Any rationales where pattern matches appeared to violate frame definitions were flagged for review.

The manual inspection confirmed that pattern assignments reflected the codebook definitions rather than superficial keyword matches. No systematic misclassification patterns were identified that would require pattern refinement. The algorithmic prevalence estimates were deemed valid for use in the frame-decomposition analysis.

#### F.4 Frame Prevalence and Exclusions

Table 4: Narrative coding codebook

| Frame                     |         | Code as 1 when rationale...                                                                                                                                                                                                                                      | Do not code as 1 when rationale...                                                                                                                                         |
|---------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Autonomy infringement     | in-     | Frames procedures as threats to individual choice, autonomy, privacy, personal freedom, consent, or procedural dignity.                                                                                                                                          | Mentions inconvenience, waiting, forms, or documentation without linking them to autonomy, privacy, dignity, or choice.                                                    |
| Procedural legitimacy     | legiti- | Frames administrative requirements as justified, reasonable, legitimate, or necessary safeguards for verification, program integrity, safety, or orderly administration.                                                                                         | Mentions documentation or registration without justifying it, or treats procedures only as obstacles.                                                                      |
| Collective responsibility | respon- | Emphasizes civic duty, public health responsibility, protection of others, community benefit, or willingness to tolerate inconvenience for collective welfare.                                                                                                   | Mentions only individual health benefits or convenience without reference to public health, community, or others.                                                          |
| Access barriers           |         | Explicitly links paperwork, waiting time, documentation, travel, scheduling, or in-person registration to reduced access, inability to comply, unequal burden, low income, work constraints, childcare, lack of regular provider access, or other vulnerability. | Merely restates that the scenario involves forms, documentation, waiting time, travel, or delay without linking these requirements to access, resources, or vulnerability. |
| Coercive backlash         |         | Mentions resentment, distrust, resistance, backlash, reactance, reduced legitimacy, or negative reaction to mandates, penalties, enforcement, or perceived coercion.                                                                                             | Describes practical inconvenience or access difficulty without resistance, distrust, resentment, or coercion.                                                              |

The access-barrier definition was intentionally restricted to avoid coding simple restatements of the high-burden prompt as a substantive frame. For example, “the process requires forms and a long wait” is not coded as access barriers unless the rationale links those requirements to reduced access or profile-specific constraints such as work flexibility, childcare, income, transportation, or lack of regular care.

Frames with little or no empirical variation are reported descriptively but excluded from the main frame-adjusted model. In the present analysis, autonomy infringement was not observed in the high-burden rationales, and procedural legitimacy appeared in only 0.1% of rationales. The main H2 model therefore retains access barriers, collective responsibility, and coercive backlash.

## G H2 Leave-One-Model-Out Robustness

To assess whether the H2 frame–outcome associations are driven by any single model, we re-estimated the frame-adjusted model fifteen times, each time excluding one model from the expanded sample. Table 5 summarizes the minimum and maximum coefficient estimates across the leave-one-model-out specifications. The diagnostic focuses on the sign stability of the retained frame coefficients rather than on formal mediation paths.

Table 5: Leave-one-model-out robustness for H2 frame coefficients

| Variable                  | Minimum coefficient | Maximum coefficient | Sign stable    |
|---------------------------|---------------------|---------------------|----------------|
| Access barriers           | 1.51                | 3.81                | Yes            |
| Collective responsibility | -1.79               | 1.62                | No             |
| Coercive backlash         | 3.52                | 11.25               | Yes            |
| PVC direct effect         | -3.64               | -0.60               | Yes (negative) |

Notes: Each row summarizes coefficients from fifteen leave-one-model-out specifications. The access-barrier and coercive-backlash coefficients are sign-stable across all specifications; collective responsibility is not. The direct PVC coefficient is negative in all fifteen specifications.

## H Frame-Manipulation Prompts and H3 Robustness

This appendix reports the prompt templates and model-specific robustness checks for the within-model frame-manipulation experiment described in the main-text section on within-model frame manipulation and analyzed in the main-text H3 results. The purpose of the experiment is to test whether governance-value frames shift predicted administrative-burden effects while holding the base model, synthetic profiles, burden scenarios, output format, and decoding settings constant.

### H.1 Frame-Manipulation Prompt Templates

The frame-manipulation experiment used four system-prompt frames. Each frame is built from the same two fixed sentences: an opening sentence and a closing instruction block. The frame clause is inserted between them, immediately after the opening sentence. The neutral frame contains no frame clause, so the neutral system prompt is byte-identical to the baseline administrative-burden system prompt reported in Appendix E (SHA256 380d779e199673cdeaf808731aad01414b60e71edba7c23a57602241ff6dc). The frame contrast is therefore measured against the true baseline condition. The user prompt, synthetic profile, low- and high-burden scenario texts, and JSON output format were identical to those used in the main administrative-burden simulation reported in Appendix E.

**Opening sentence:** You are estimating how likely the person described below would be to get vaccinated under the specified access conditions.

**Closing instruction block:** Base the estimate only on the information provided in the profile and the scenario; do not add assumptions beyond them. Respond ONLY with a single JSON object and nothing else. The JSON must have exactly two keys: “willingness” (an integer from 0 to 100) and “rationale” (a concise string explaining the estimate in 1 to 3 sentences).

The four frame conditions differ only in the inserted clause:

- **Neutral frame:** no clause inserted. Full system prompt SHA256: 380d779e199673cdeaf808731aad01414b60
- **Autonomy frame:** “When forming the estimate, give special weight to individual choice, privacy, personal autonomy, and procedural dignity.” Full system prompt SHA256: 8b7d948e661603aa989881
- **Collective-obligation frame:** “When forming the estimate, give special weight to civic responsibility, public health coordination, and willingness to tolerate reasonable inconvenience for collective protection.” Full system prompt SHA256: fe7512cde053405edf59cbaaf6d1402f68d69e7c47898
- **Equity/access frame:** “When forming the estimate, give special weight to unequal procedural costs, access barriers, work constraints, childcare constraints, and burdens faced by disadvantaged groups.” Full system prompt SHA256: cba0156ababefd86611eb267ebf9ccddb1d60fe5a8cecf4e289

For each model  $m$ , frame condition  $c$ , profile  $i$ , and repetition  $r$ , the model evaluated both the low-burden and high-burden vaccination-access scenarios in fresh conversation contexts. The frame-specific burden effect was computed as:

$$\Delta_{mcir} = W_{mcir}^{low} - W_{mcir}^{high}.$$

Positive values indicate a larger predicted reduction in vaccination willingness under the high-burden condition.

## H.2 Mean Burden Effects by Model and Frame

Table 6 reports mean burden effects by model and frame condition for the four-model H3 subset. The descriptive means show that the collective-obligation frame lowers simulated burden effects in all four models, and that the autonomy frame increases simulated burden effects in three of the four models. The remaining contrasts are model-specific rather than uniform: for Qwen3.7 Plus the autonomy frame lowers the mean burden effect (53.71 vs. 56.59 under the neutral frame), and the equity/access frame has little effect for that model (55.64) and for Llama 3.1 70B Instruct (36.42 vs. 27.11 under neutral, a positive but smaller shift than the autonomy frame). The absolute level of the burden effect also differs substantially across models under the neutral frame (25.59 for GPT-4.1 to 56.59 for Qwen3.7 Plus), so frame effects are interpreted relative to each model’s own neutral baseline.

Mistral Large is a particularly informative robustness case. It has the lowest PVC score in the expanded fifteen-model sample, and all three frame manipulations shift its simulated burden effects in the expected directions. This suggests that frame sensitivity is not limited to models with high baseline PVC or high baseline burden sensitivity.

Table 6: Mean burden effects by model and frame

| Model                  | Frame                 | Mean $\Delta$ | SD    | $N$ |
|------------------------|-----------------------|---------------|-------|-----|
| GPT-4.1                | Neutral               | 25.59         | 8.42  | 135 |
| GPT-4.1                | Autonomy              | 34.15         | 6.84  | 135 |
| GPT-4.1                | Collective obligation | 18.62         | 6.22  | 135 |
| GPT-4.1                | Equity/access         | 34.00         | 9.46  | 135 |
| Llama 3.1 70B Instruct | Neutral               | 27.11         | 9.69  | 135 |
| Llama 3.1 70B Instruct | Autonomy              | 37.19         | 7.98  | 135 |
| Llama 3.1 70B Instruct | Collective obligation | 25.33         | 10.42 | 135 |
| Llama 3.1 70B Instruct | Equity/access         | 36.42         | 12.90 | 134 |
| Qwen3.7 Plus           | Neutral               | 56.59         | 7.64  | 135 |
| Qwen3.7 Plus           | Autonomy              | 53.71         | 6.54  | 135 |
| Qwen3.7 Plus           | Collective obligation | 41.93         | 11.34 | 135 |
| Qwen3.7 Plus           | Equity/access         | 55.64         | 6.91  | 135 |
| Mistral Large          | Neutral               | 37.04         | 6.44  | 135 |
| Mistral Large          | Autonomy              | 41.11         | 5.38  | 135 |
| Mistral Large          | Collective obligation | 16.93         | 7.53  | 135 |
| Mistral Large          | Equity/access         | 39.63         | 6.95  | 135 |

Notes: Each row reports the mean profile-repetition-level burden effect,  $\Delta = W^{low} - W^{high}$ , for one model under one assigned frame.  $N = 135$  for each model-frame cell, corresponding to 27 profiles and five repetitions, except where a response failed the structured-output checks: Llama 3.1 70B Instruct equity/access ( $N = 134$ ), and Qwen3.7 Plus ( $N = 114, 104, 108$ , and  $120$  across the neutral, autonomy, collective-obligation, and equity/access frames respectively, reflecting calls rejected by the serving endpoint). Positive values indicate larger predicted reductions in willingness under the high-burden condition.

### H.3 Model-Specific Frame-Manipulation Effects

Table 7 reports model-specific frame regressions for the four-model H3 subset. Each model-specific regression estimates  $\Delta$  as a function of frame indicators and profile fixed effects, using the neutral frame as the reference condition. The pooled column corresponds to the main H3 specification reported in the main text, which pools the same four models and includes model and profile fixed effects.

Table 7: Model-specific frame-manipulation effects

| Frame contrast                    | GPT-4.1  | Llama 3.1 70B | Qwen3.7 Plus | Mistral Large | Pooled    |
|-----------------------------------|----------|---------------|--------------|---------------|-----------|
| Autonomy vs. neutral              | 8.56***  | 10.07***      | -2.87**      | 4.07***       | 4.96***   |
| Collective obligation vs. neutral | -6.97*** | -1.78         | -14.65***    | -20.11***     | -10.88*** |
| Equity/access vs. neutral         | 8.41***  | 9.36***       | -0.94        | 2.59*         | 4.84***   |

Notes: Each model-specific column reports a regression of  $\Delta$  on frame indicators and profile fixed effects for the indicated model. The pooled column reports the main H3 specification for the four-model H3 subset with model and profile fixed effects. Standard errors are clustered at the model-profile level in all columns. The neutral frame is the omitted reference category. Positive coefficients indicate larger simulated burden effects relative to neutral; negative coefficients indicate smaller simulated burden effects. Standard errors and 95% confidence intervals are reported in Table 8. \* $p < 0.05$ , \*\*\* $p < 0.001$ .

Table 8 reports the corresponding standard errors and confidence intervals. The model-specific results show both robustness and heterogeneity within the H3 subset. The collective-obligation frame reduces simulated burden effects in all four models (significantly in GPT-4.1, Qwen3.7 Plus, and Mistral Large; not statistically distinguishable from zero in Llama 3.1 70B Instruct). The autonomy frame increases simulated burden effects in GPT-4.1, Llama 3.1 70B Instruct, and Mistral Large, but *reverses sign* for Qwen3.7 Plus ( $-2.87$ , 95% CI  $[-4.53, -1.22]$ ), where activating autonomy concerns reduces the simulated burden effect rather than amplifying it. The equity/access frame is positive and significant for GPT-4.1, Llama 3.1 70B Instruct, and Mistral Large, and statistically indistinguishable from zero for Qwen3.7 Plus. Mistral Large is especially informative because it has the lowest PVC score in the expanded fifteen-model sample, yet all three frame manipulations shift its simulated burden effects in the expected directions. These results support the pooled H3 finding for the collective-obligation contrast while showing that model endpoints differ in both the magnitude and the sign of responsiveness to specific governance-value frames; the pooled autonomy and equity estimates are averages over a genuinely mixed set of model-level responses, and the Qwen3.7 Plus reversal is the clearest caution against reading a pooled frame effect as a property of LLMs in general.

## I Additional Robustness Checks

This appendix reports additional robustness checks for the H1 orientation-effect analysis. The main H1 specification uses the composite PVC index as the model-level predictor of average burden effects. Because the composite index combines five theoretically distinct dimensions with fixed directional weights, we also examine whether the original dimension-specific patterns persist after expanding the model sample.

Table 8: Model-specific frame-manipulation effects with standard errors and confidence intervals

| Model                  | Frame contrast                    | Coefficient | SE   | 95% CI           |
|------------------------|-----------------------------------|-------------|------|------------------|
| GPT-4.1                | Autonomy vs. neutral              | 8.56***     | 0.76 | [6.99, 10.12]    |
| GPT-4.1                | Collective obligation vs. neutral | -6.97***    | 0.81 | [-8.63, -5.31]   |
| GPT-4.1                | Equity/access vs. neutral         | 8.41***     | 0.77 | [6.83, 9.99]     |
| Llama 3.1 70B Instruct | Autonomy vs. neutral              | 10.07***    | 1.46 | [7.07, 13.08]    |
| Llama 3.1 70B Instruct | Collective obligation vs. neutral | -1.78       | 1.21 | [-4.27, 0.72]    |
| Llama 3.1 70B Instruct | Equity/access vs. neutral         | 9.36***     | 1.75 | [5.76, 12.97]    |
| Qwen3.7 Plus           | Autonomy vs. neutral              | -2.87**     | 0.81 | [-4.53, -1.22]   |
| Qwen3.7 Plus           | Collective obligation vs. neutral | -14.65***   | 1.06 | [-16.84, -12.47] |
| Qwen3.7 Plus           | Equity/access vs. neutral         | -0.94       | 0.84 | [-2.66, 0.78]    |
| Mistral Large          | Autonomy vs. neutral              | 4.07***     | 0.98 | [2.05, 6.09]     |
| Mistral Large          | Collective obligation vs. neutral | -20.11***   | 0.89 | [-21.93, -18.29] |
| Mistral Large          | Equity/access vs. neutral         | 2.59*       | 1.13 | [0.28, 4.91]     |
| Pooled                 | Autonomy vs. neutral              | 4.96***     | 0.70 | [3.57, 6.35]     |
| Pooled                 | Collective obligation vs. neutral | -10.88***   | 0.84 | [-12.54, -9.21]  |
| Pooled                 | Equity/access vs. neutral         | 4.84***     | 0.70 | [3.45, 6.24]     |

Notes: Each row reports a frame contrast relative to the neutral frame. Model-specific regressions include profile fixed effects. The pooled regression includes model and profile fixed effects. Standard errors are clustered at the model–profile level (108 clusters) because profiles are the controlled test cases and queries within a profile are repeated measures. The pooled specification uses  $N = 2,159$  profile–repetition observations (540 for GPT-4.1, 539 for Llama 3.1 70B Instruct, 540 for Qwen3.7 Plus, and 540 for Mistral Large, after excluding responses that failed the structured-output checks). The dependent variable is  $\Delta = W^{low} - W^{high}$ . Positive coefficients indicate larger simulated burden effects relative to neutral. Significance stars are based on Holm–Bonferroni-adjusted  $p$ -values:

\* $p < 0.05$ , \*\*\* $p < 0.001$ .

## I.1 Expanded-Sample H1 Robustness

Table 9 summarizes the key H1 robustness results for the original nine-model sample and the expanded fifteen-model sample. The composite PVC index does not predict model-level burden effects in either sample. In the original nine-model sample, the PVC coefficient is negative and statistically indistinguishable from zero ( $\hat{\beta} = -4.24$ ,  $p = 0.219$ ,  $R^2 = 0.206$ ). After expanding to fifteen endpoints, the PVC coefficient remains weak and statistically indistinguishable from a null effect at conventional power ( $\hat{\beta} = -2.14$ ,  $R^2 = 0.090$ ).

The expanded sample also changes the interpretation of the dimension-specific collective-obligation result. In the original nine-model sample, collective obligation (CO) was positively associated with model-level burden effects but did not reach conventional significance ( $\hat{\beta} = 10.81$ ,  $p = 0.179$ ,  $R^2 = 0.242$ ). After the six OpenRouter-accessed models are added, the association shrinks and remains non-significant ( $\hat{\beta} = 1.28$ ,  $p = 0.725$ ,  $R^2 = 0.010$ ). This indicates that the CO association is sensitive to model composition and should be interpreted as an exploratory signal at best rather than as a stable dimension-level predictor.

The expanded-sample measurement diagnostics also suggest caution in interpreting the composite PVC formula. CO is the only dimension with acceptable exploratory internal consistency, but removing CO from the composite does not change the substantive conclusion: the leave-CO-out PVC coefficient remains statistically indistinguishable from zero (Table 9). This does not validate an alternative index; the composite association is weak under either weighting.

Table 9: H1 robustness across original and expanded model samples

| Predictor and sample            | Coefficient | $p$   | $R^2$ | Interpretation  |
|---------------------------------|-------------|-------|-------|-----------------|
| PVC composite, $N = 9$          | -4.24       | 0.219 | 0.206 | Null            |
| PVC composite, $N = 15$         | -2.14       | 0.278 | 0.090 | Null            |
| Collective obligation, $N = 9$  | 10.81       | 0.179 | 0.242 | Not significant |
| Collective obligation, $N = 15$ | 1.28        | 0.725 | 0.010 | Not stable      |
| Leave-CO-out PVC, $N = 15$      | -2.22       | 0.333 | 0.072 | Null            |

Notes: Each row reports a separate bivariate model-level regression of  $\bar{\Delta}_m$  on the listed predictor or PVC variant. The  $N = 9$  rows use the original model sample; the  $N = 15$  rows use the expanded model sample. Higher  $\bar{\Delta}_m$  indicates a larger predicted reduction in willingness under high administrative burden. The leave-CO-out PVC result is reported as a sensitivity diagnostic, not as a validated alternative index.

## I.2 Leave-One-Model-Out Diagnostics

Leave-one-model-out diagnostics further support the main H1 conclusion. The composite PVC coefficient is not a stable predictor of model-level burden effects. In the expanded sample, the bivariate PVC association remains weak and statistically indistinguishable from zero. The CO pattern is not significant in either sample: the coefficient is positive but small in the nine-model analysis and shrinks further after adding six additional endpoints.

These diagnostics reinforce the interpretation that H1 does not provide evidence for a stable aggregate orientation-effect relationship. The PVC index is useful as a theoretically motivated diagnostic construct, but the expanded analysis does not support treating it as a validated scalar predictor of simulated burden-effect magnitude.

### I.3 Power and Minimum Detectable Effect Diagnostics

The original nine-model H1 analysis was underpowered for small or moderate model-level associations. In that sample, the approximate 80% minimum detectable slope for the composite PVC index was  $\beta \approx 10.2$  (5.7 in the fifteen-endpoint sample). The observed PVC coefficient was far smaller than this threshold. Expanding the model sample to fifteen endpoints improves coverage but does not eliminate the basic limitation that H1 relies on model-level variation.

For this reason, the H1 estimates should be interpreted as descriptive diagnostics rather than high-powered population estimates. The expanded results are most informative as evidence against a large linear association between the composite PVC index and simulated burden effects in this model sample. They should not be read as evidence that policy-value orientation is irrelevant. Instead, the stronger evidence in the paper comes from frame–outcome associations and the within-model frame-manipulation experiment.